



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECI 1143

PROBABILITY AND STATISTICAL DATA ANALYSIS

PROJECT 1

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SEMESTER II (2023/2024)

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1.0 INTRODUCTION

Nowadays, many students lead busy lives with demanding schedules. Sometimes, making time for a healthy, reasonably priced diet might be difficult for them. Following this, students frequently choose for quick and easy meal options like fast food or prepackaged snacks which might not necessarily be the healthiest one. However, students have more alternatives for where and what to eat because of the growth of meal delivery services and on-campus eating options.

Understanding how they manage their eating and spending is crucial because food consumes a significant portion of their budget. In order to gather data on the number of meals taken daily and the amount of money spent on it, we surveyed our sample of Universiti Teknologi Mara (UTM) and other university students in Malaysia. This study is to gather data from Malaysia's university students in order to observe their eating and dietary habits. From the survey's questions, we may observe and form conclusions about how much students spend on meals per day and how often they eat whether they prefer to eat a home-cooked meal, eating out or takeout. Additionally, we can see how concerned students have about their budget and the nutrition they take per meal.

In this report, we will use a variety of graphs that represent the data based on the information we collected from the survey and also can get the statistics of meals taken per day and how much students spent on it.

2.0 DATA COLLECTION

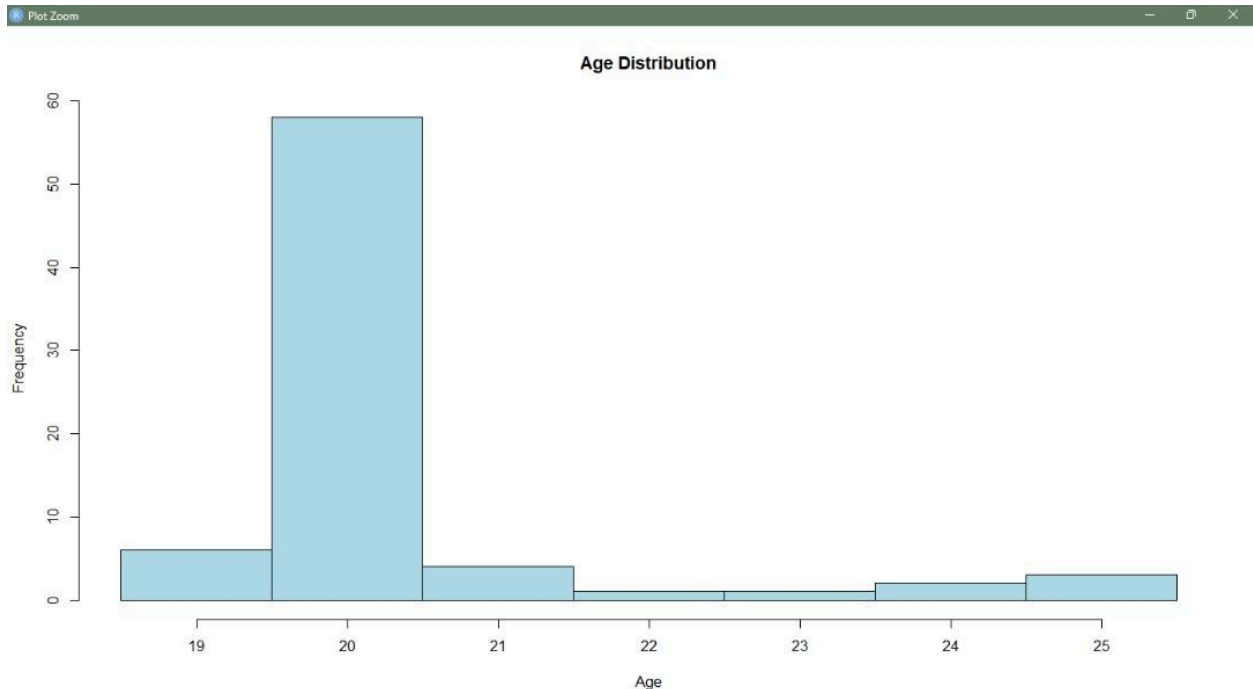
Population : Students

Sample : University Students

Variable	Type of Variable	Level of Measurement	Representation
Age	Quantitative	Ratio	Histogram, Frequency Distribution
Gender	Categorical	Nominal	Pie Chart
Student Status	Categorical	Nominal	Pie Chart
Meals per day	Quantitative	Ratio	Dot Plot, Box plot
Source for Meal	Categorical	Nominal	Bar Chart
Dining out	Quantitative	Ratio	Scatter plot, Box Plot
Frequency of Breakfast Consumption	Quantitative	Ordinal	Dot Plot, Frequency Distribution
Frequency of Lunch Consumption	Quantitative	Ordinal	Dot Plot, Frequency Distribution
Frequency of Dinner Consumption	Quantitative	Ordinal	Dot Plot, Frequency Distribution
Money Spend on Breakfast	Quantitative	Ratio	Histogram, Frequency Distribution
Money Spend on Lunch	Quantitative	Ratio	Histogram, Frequency Distribution
Money Spend on Dinner	Quantitative	Ratio	Histogram, Frequency Distribution
Importance on Budgeting Food Expenses	Qualitative	Ordinal	Bar Chart, Box Plot
Importance of Nutrition in Meal Choices	Qualitative	Ordinal	Pie Chart, Box Plot
Money Spend on Food per Day	Quantitative	Ratio	Dot Plot, Box Plot
Money Spend on Food Per Month	Quantitative	Ratio	Dot Plot, Box Plot
Allocated Money for Food Expenses	Quantitative	Ratio	Pie Chart, Box Plot

3.0 DATA ANALYSIS

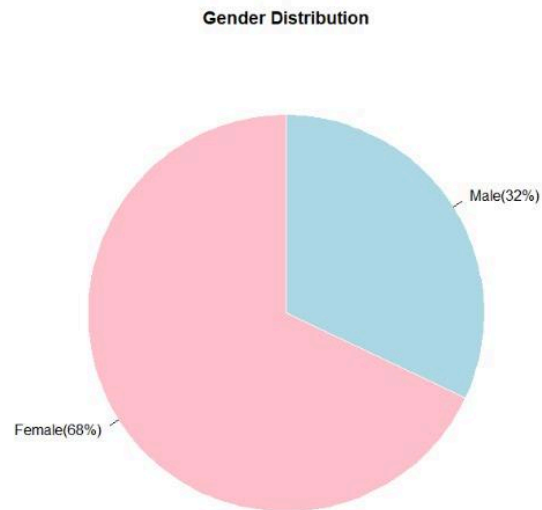
1. Age



```
> # Display the frequency distribution table with percentages
> print(frequency_distribution)
  Age Frequency Percentage Relative_Frequency
1  19         6    8.000000         0.08000000
2  20        58   77.333333         0.77333333
3  21         4    5.333333         0.05333333
4  22         1    1.333333         0.01333333
5  23         1    1.333333         0.01333333
6  24         2    2.666667         0.02666667
7  25         3    4.000000         0.04000000
```

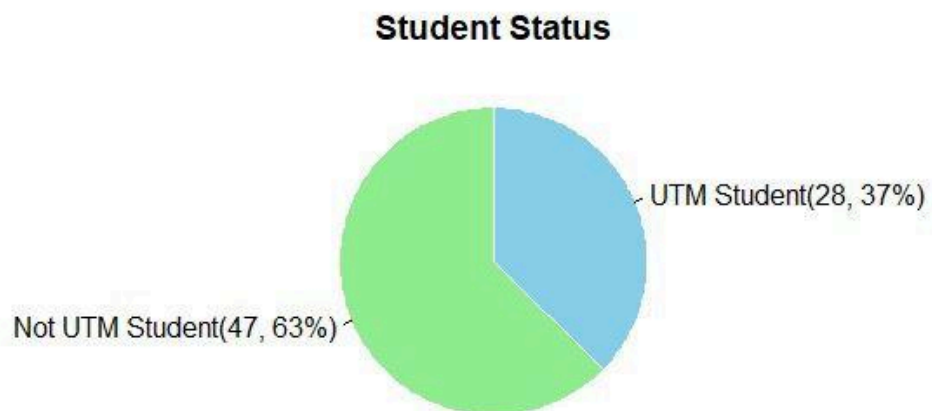
Explanation : In this survey, the respondents are required to fill in their age. From this graph, we can see that the majority, 77.33% of the students are 20 years old. Then 8.00% of the students are 19 years old. 21 year old students are 5.33%. The second lowest is 4.00% who are 25 years old. Followed by the lowest number of results with 1.33% which are 22 and 23 years old students.

2. Gender



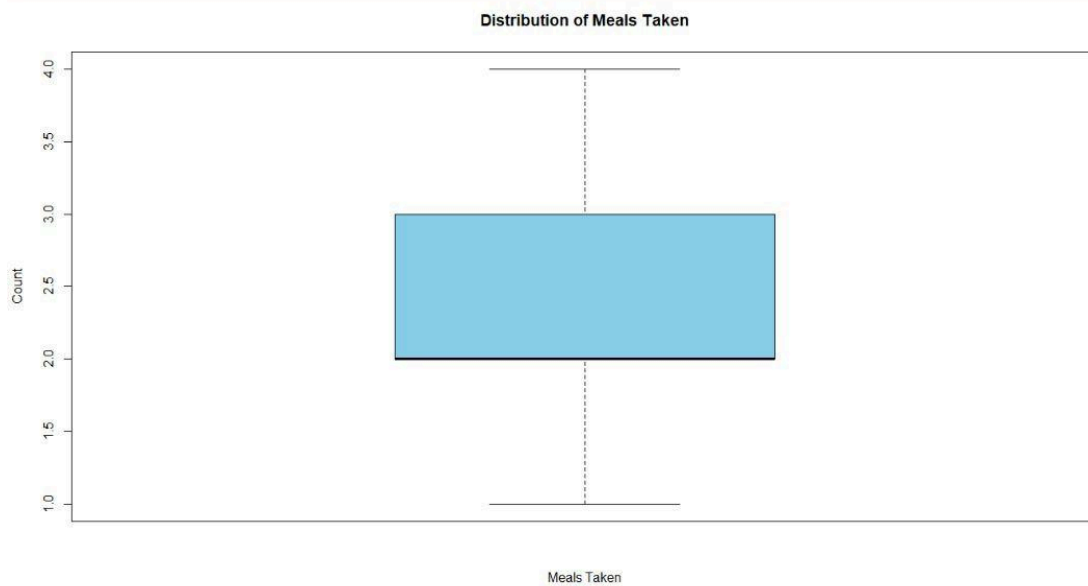
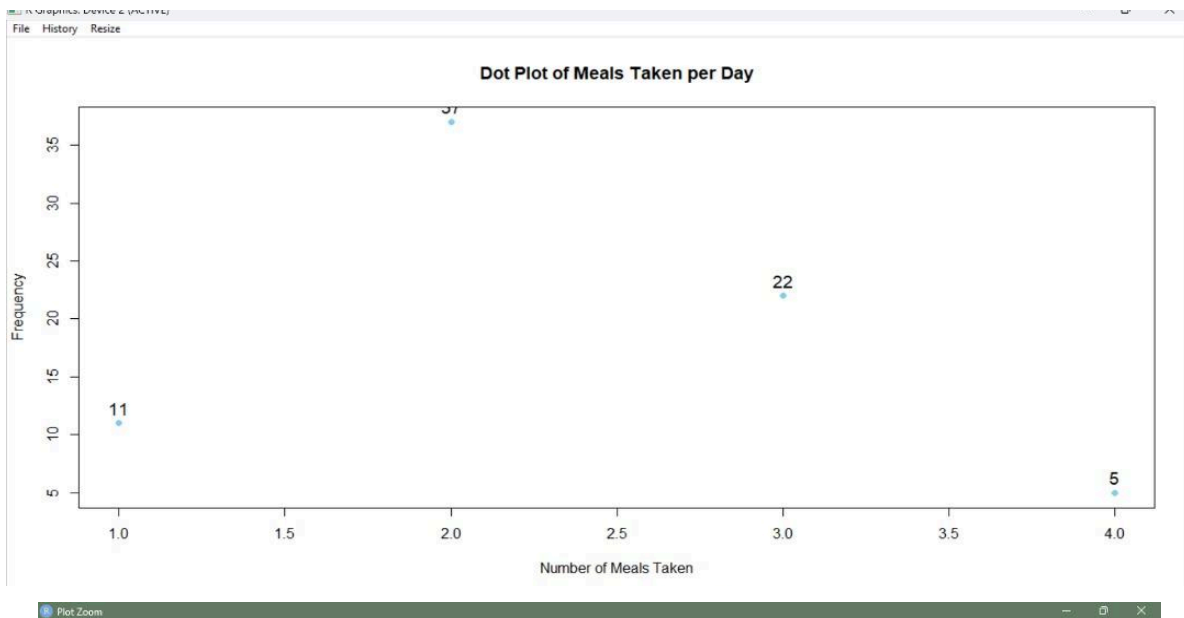
Explanation: In this survey, there are 68% female and 32% male of students' response.

3. Student status



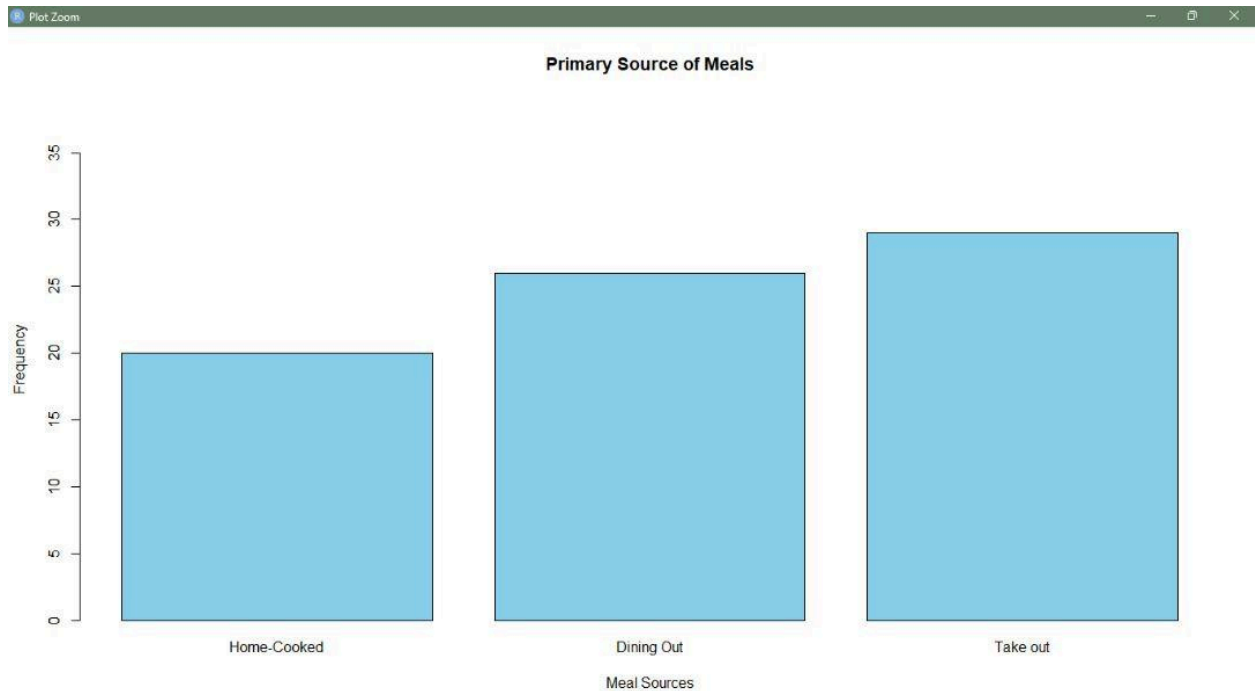
Explanation : For this section we can see that the majority of the students are 47.63% who are not UTM students. While students who are studying in UTM are 28.37%.

4. Meals per day



Explanation: There are 37% of students taking meals twice a day while 22 students are taking 3 meals per day and the least number of students with 11 students are taking 1 meal per day.

5. Source of Meals



Explanation: In this section, students pick on how they preferably eat their meals. From this graph, we can see that the majority of students prefer to take out from restaurants, followed by dining out. Lastly, the least number of picks is home-cooked meals.

6. Dining out



Explanation: From the data, we found that the majority of the students dining out 1-2 times a week with an amount of 32 students. followed by the second most time which is more than 4 times a week by 24 students, 3-4 times a week by 17 students. Among those times, there are only 2 students who are not dining out during the week.

7. Breakfast Consumption

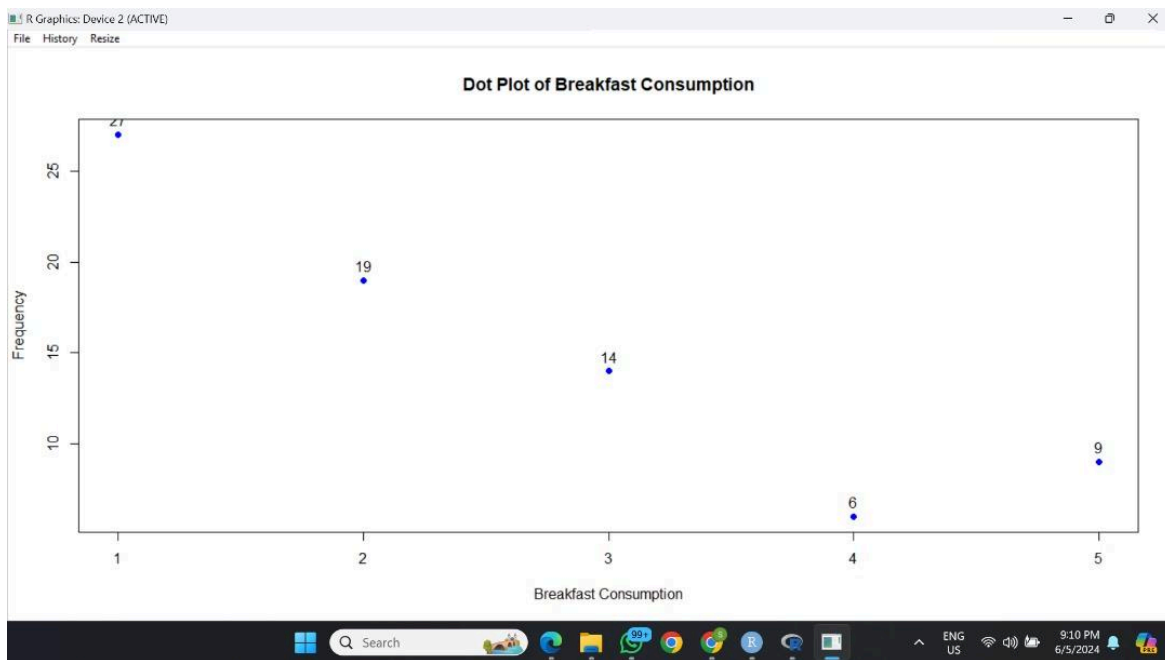


Table: Frequency Distribution of Breakfast Consumption Ratings

Rating	Frequency	Percentage	RelativeFrequency
1	27	36.00000	36.00000
2	19	25.33333	25.33333
3	14	18.66667	18.66667
4	6	8.00000	8.00000
5	9	12.00000	12.00000

Explanation: From the data, we found that 36% of students rate their breakfast consumption with 1, 25.33% of students rated 2, 18.67% of students rated 3, 12% of students rated 5 and 8% of students rated 4. Our data for breakfast consumption is represented by dot plots.

8. Lunch Consumption

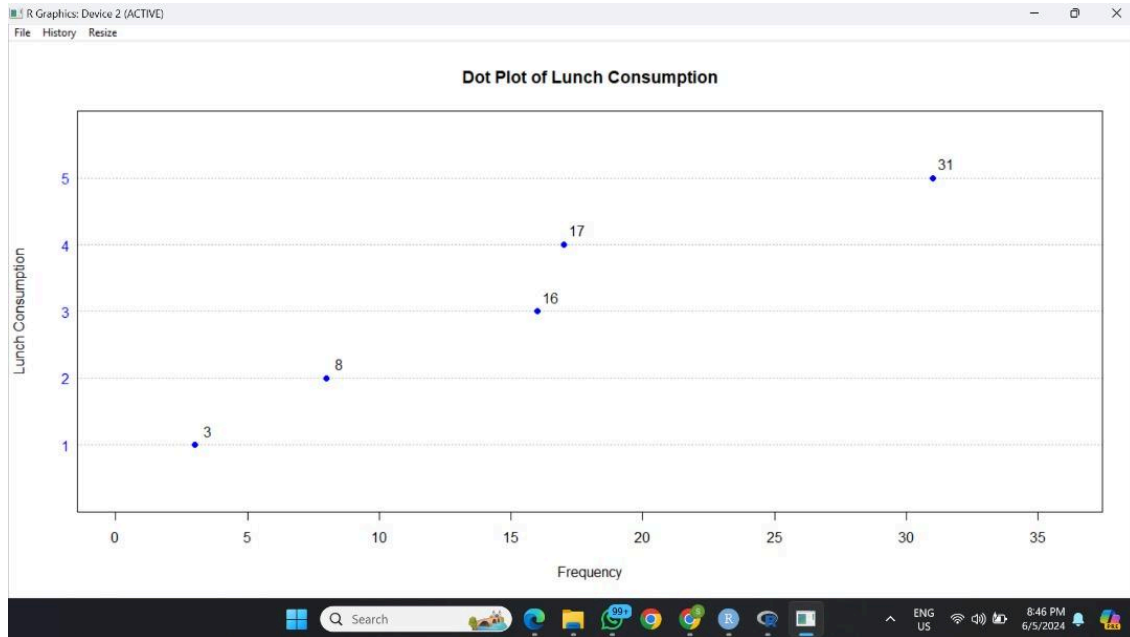


Table: Frequency Distribution of Lunch Consumption (Scale 1 to 5)

scale	Frequency	Percentage	RelativeFrequency
1	3	4.00000	4.00000
2	8	10.66667	10.66667
3	16	21.33333	21.33333
4	17	22.66667	22.66667
5	31	41.33333	41.33333

Explanation: In this section, the respondents are required to score which scale on how they would rate the frequency of their lunch consumption. From this graph, we can see that the majority, 41.33% of the students consume frequently. 22.67% of the students score 3 which is in the middle, while 21.33% less frequent, 10.67% is the second lowest and only 24% consume the lowest.

9. Dinner Consumption

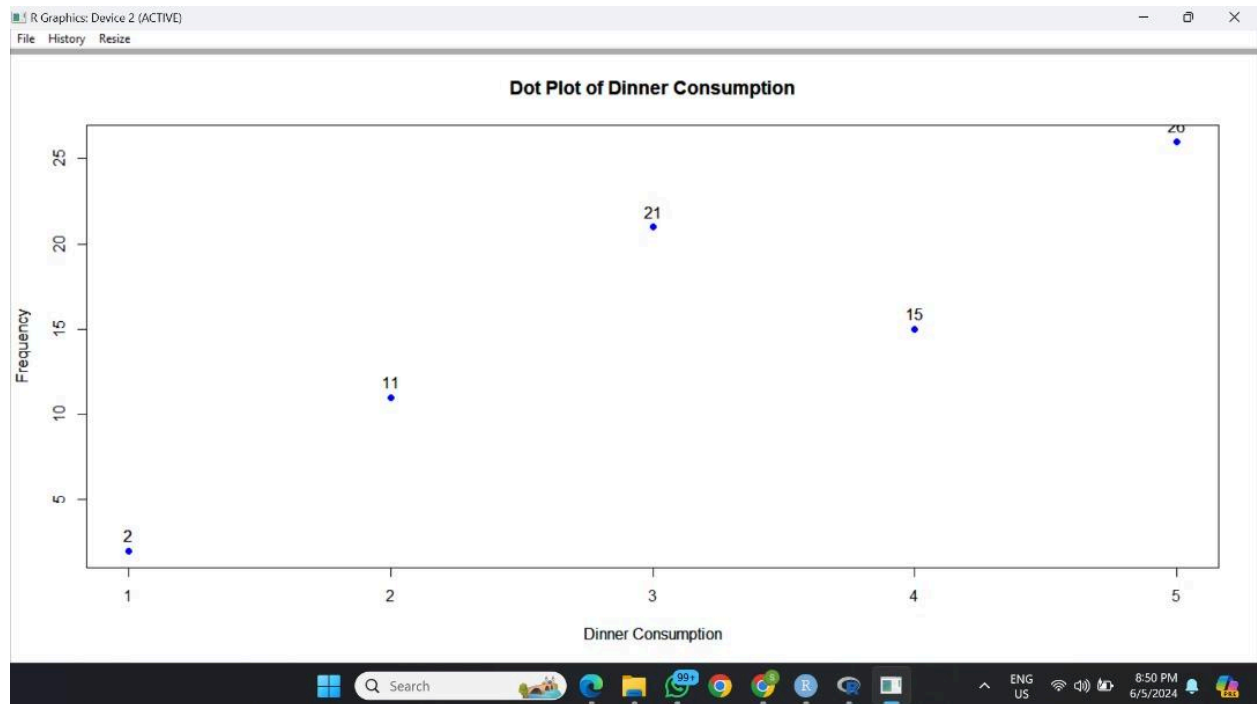


Table: Frequency Distribution of Dinner Consumption (Scale 1 to 5)

Scale	Frequency	Percentage	RelativeFrequency
1	2	2.666667	2.666667
2	11	14.666667	14.666667
3	21	28.000000	28.000000
4	15	20.000000	20.000000
5	26	34.666667	34.666667

Explanation : In this section, the respondents are required to score which scale on how they would rate the frequency of their dinner consumption. From this graph, we can see that the majority, 34.67% of the students consume frequently. 28% of the students score 3 which is in the middle, while 20% less frequent, 14.67% is the second lowest and only 2.67% consume the lowest.

10. Breakfast expenses

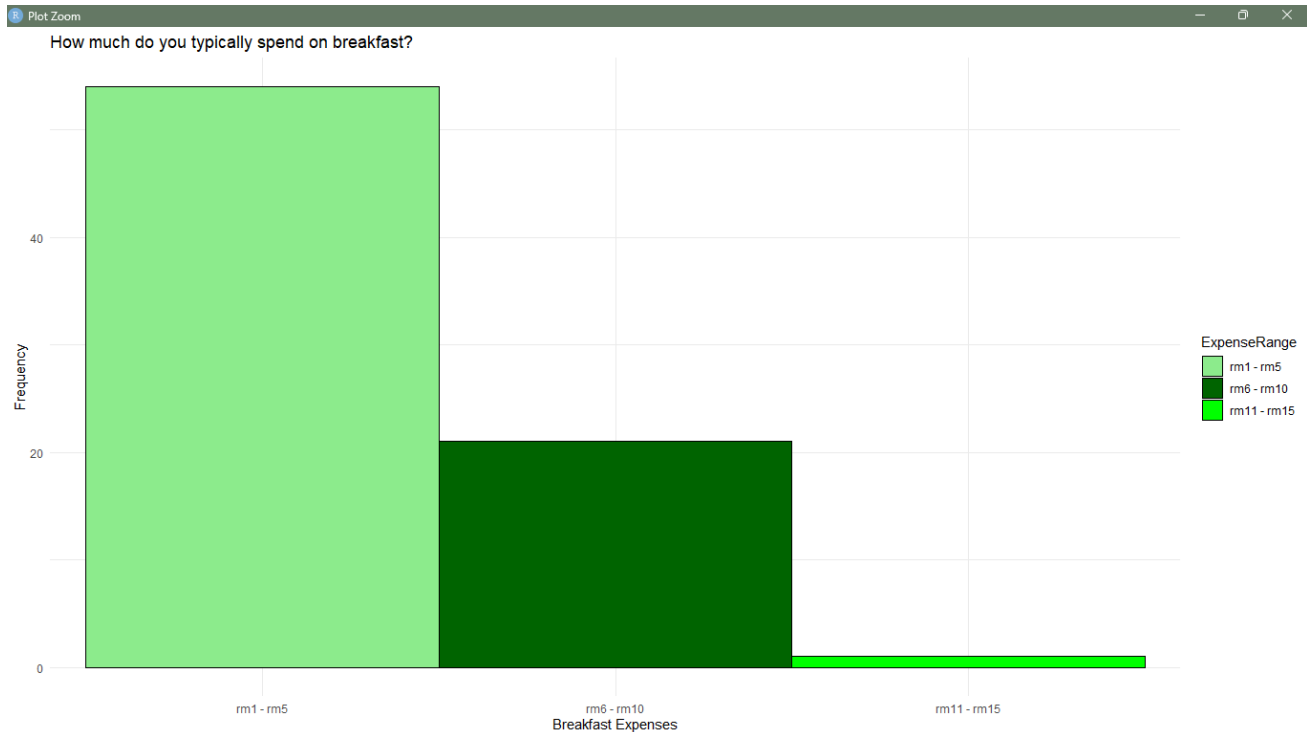


Table: Frequency Distribution of Breakfast Expenses

ExpenseRange	Frequency	Percentage	RelativeFrequency
rm1-5	54	71.052632	71.052632
rm6-10	21	27.631579	27.631579
rm11-15	1	1.315789	1.315789

Explanation : In this section, the respondents are required to tick on how much money they spent for breakfast. From this graph, we can see that the majority, 71.05% of the students spent only RM1-RM5 for their breakfast. Spending RM6-RM10 for breakfast is the second highest result of the student with 27.63%. Followed by the lowest number of results with 1.32% which means not many students were spending money for breakfast more than RM10.

11. Lunch expenses

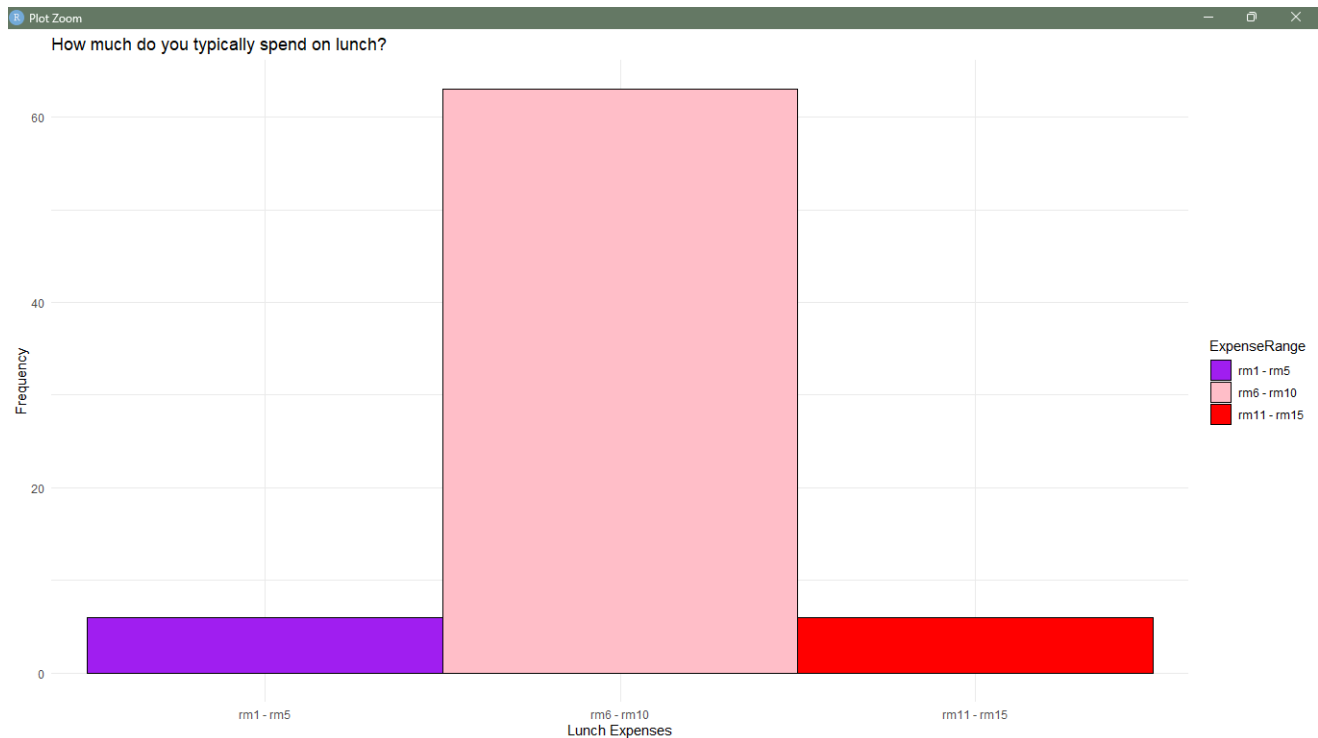


Table: Frequency Distribution of Lunch Expenses

ExpenseRange	Frequency	Percentage	RelativeFrequency
rm1-5	6	8	8
rm6-10	63	84	84
rm11-15	6	8	8

Explanation: From the data, we found out that most students spent RM6-RM10 for their lunch which is 84% followed by students spending RM1-RM5 and RM11-RM15 equally with a percentage of 8%.

12. Dinner expenses

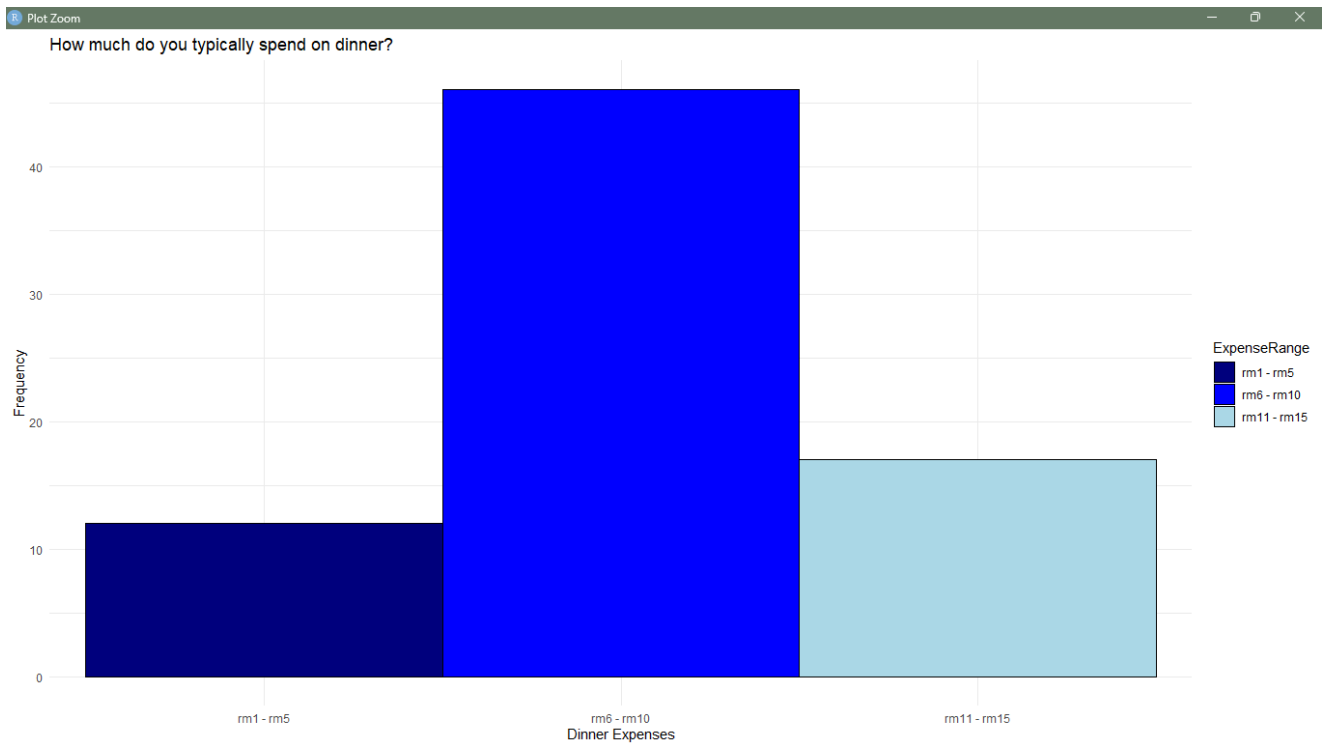
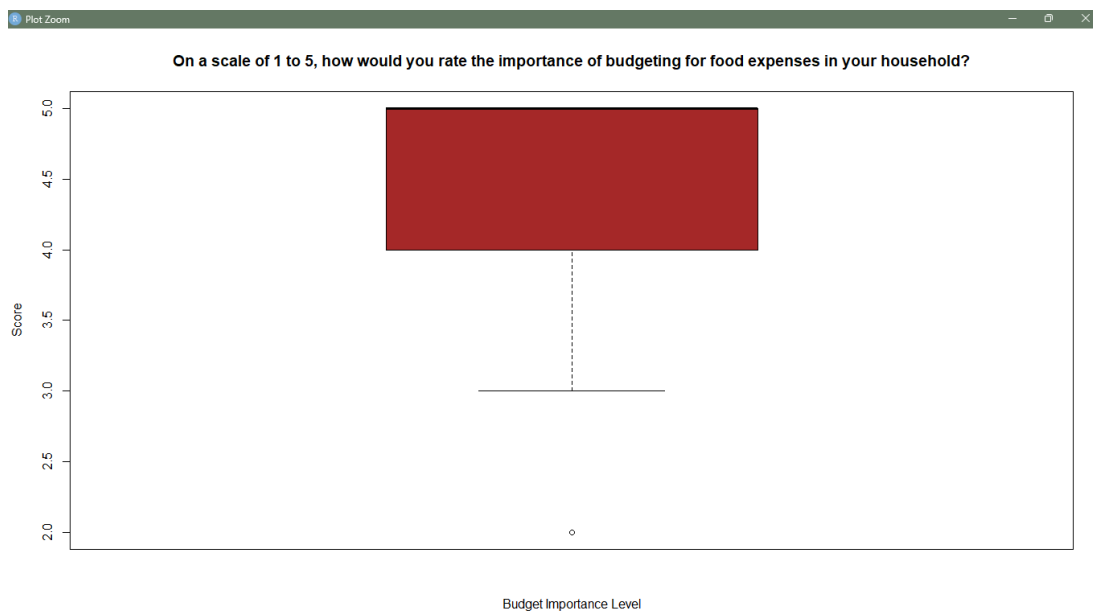
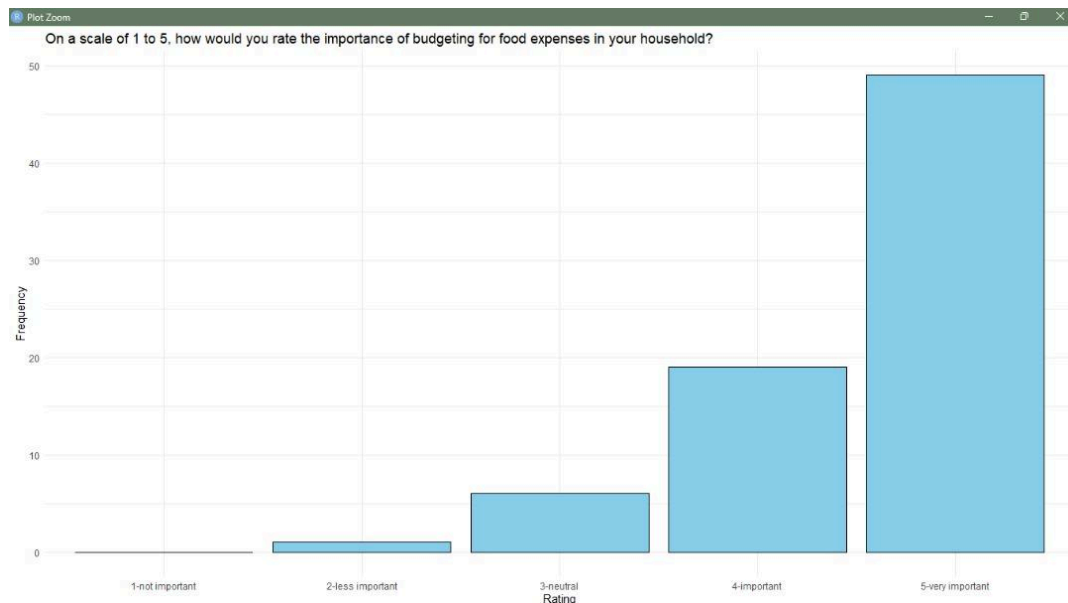


Table: Frequency Distribution of Dinner Expenses

ExpenseRange	Frequency	Percentage	RelativeFrequency
rm1-5	12	16.00000	16.00000
rm6-10	46	61.33333	61.33333
rm11-15	17	22.66667	22.66667

Explanation: From the data, we found that 61.33% of the students have spent RM6-RM10 for their dinner per day, 22.67% of the students have spent RM11-RM15 for dinner per day and 16% of the students have spent RM1-RM5 for dinner per day. Our data for dinner expenses is being represented via histogram.

13. Importance for budgeting food expenses

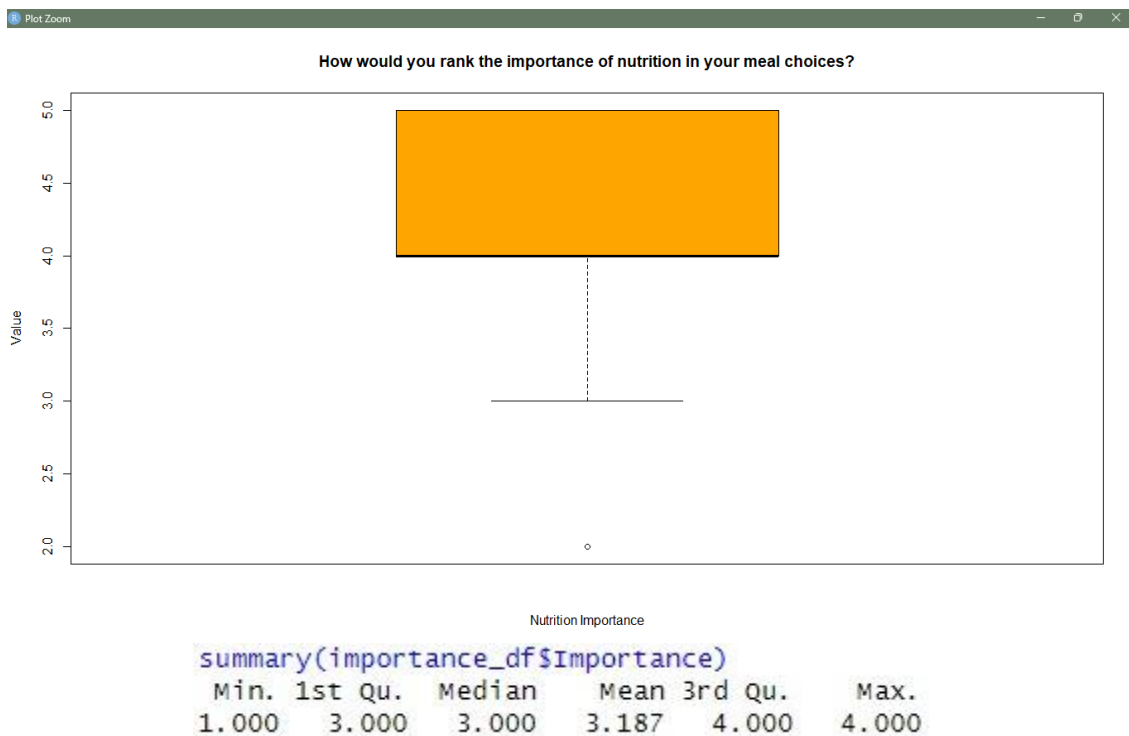
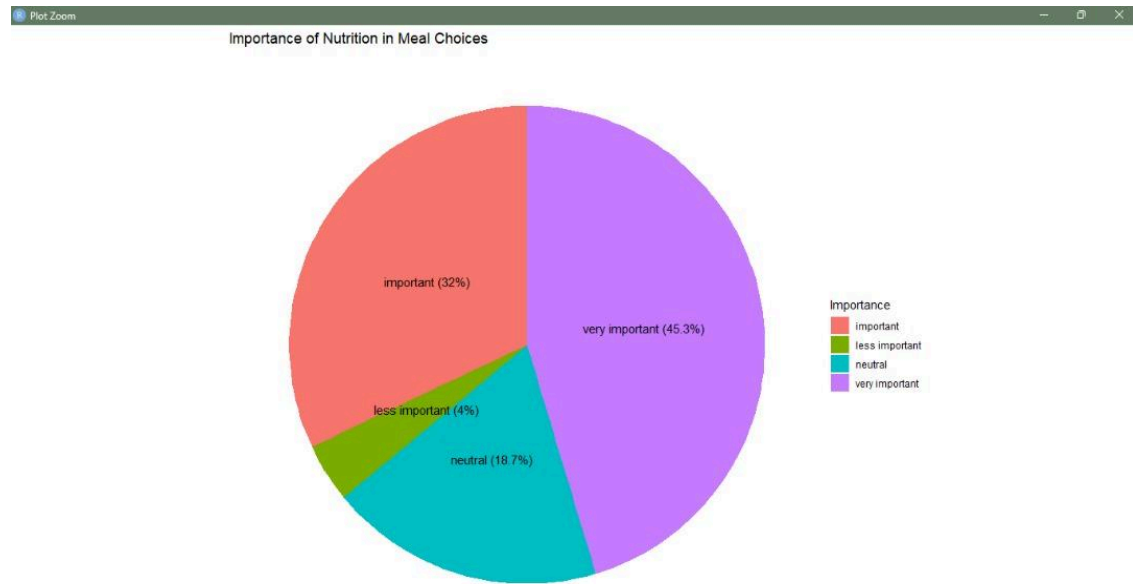


```
summary(importance_df$Importance)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 1.000  3.000   3.000   3.187  4.000   4.000
```

Explanation: During the survey, students have to pick one of the importance of budgeting for food expenses in their household on a scale of 1-5. From the data, we can see that almost all of the students rate 5 for the importance of budgeting for food which means they know that it is very crucial to spend some money for their daily meal. While the others students rated the importance of budget on

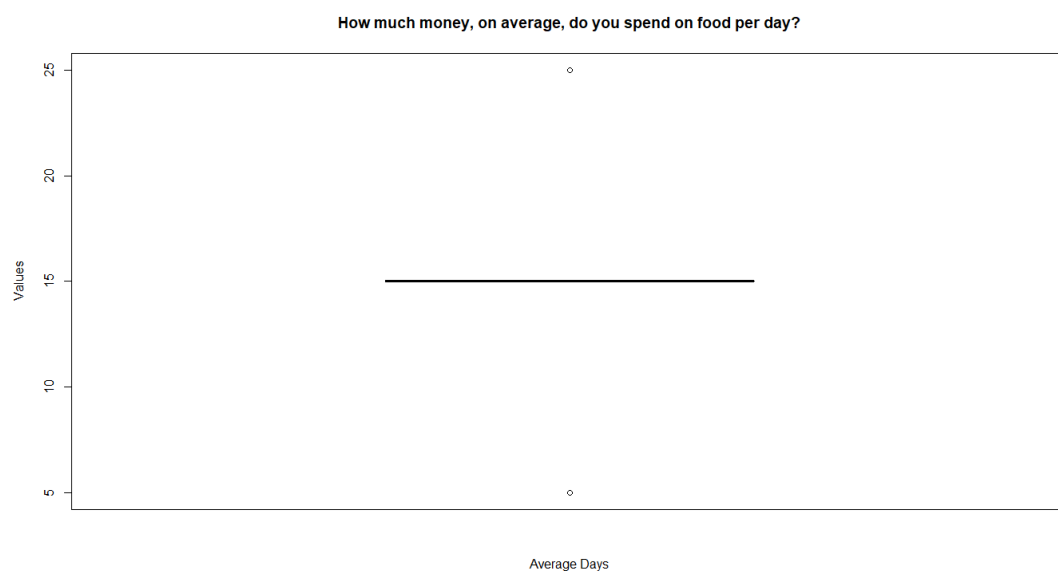
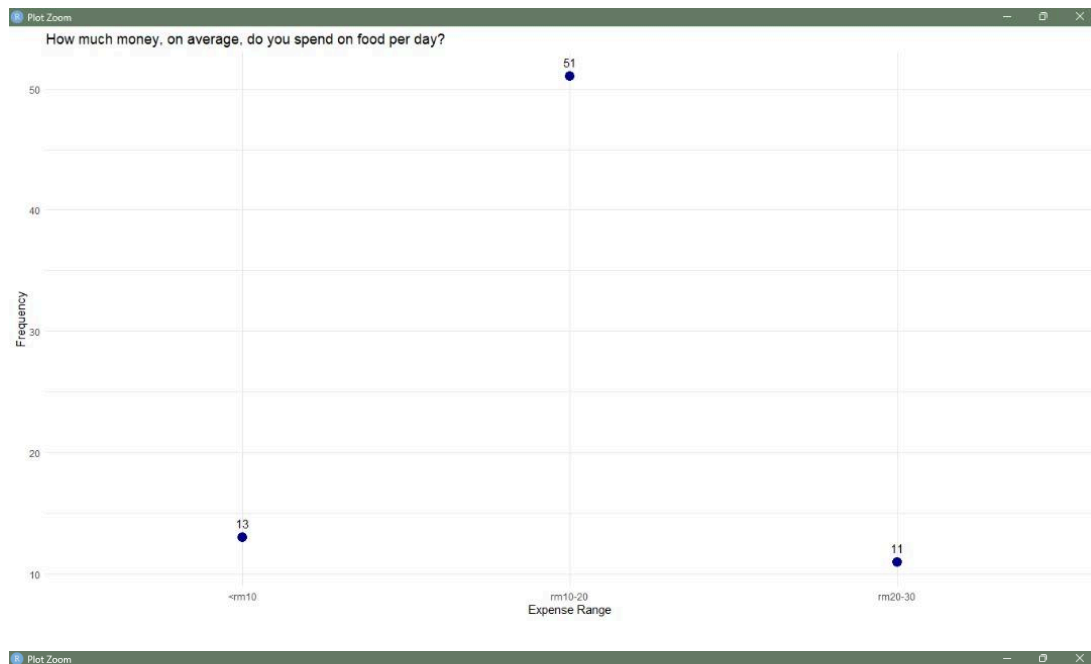
food with 4, 3, 2 and 1 respectively. This shows that most of the students know the importance of having a budget for foods while maintaining a healthy lifestyle.

14. Importance of nutrition



Explanation: From the data, 45.3% of the students choose the importance of nutrition in meal choices as very important and crucial, while 32% of the students choose important, 18.7% of the students choose neutral and lastly 4% of the students choose less important. We represented this data using a pie chart as well as boxplot.

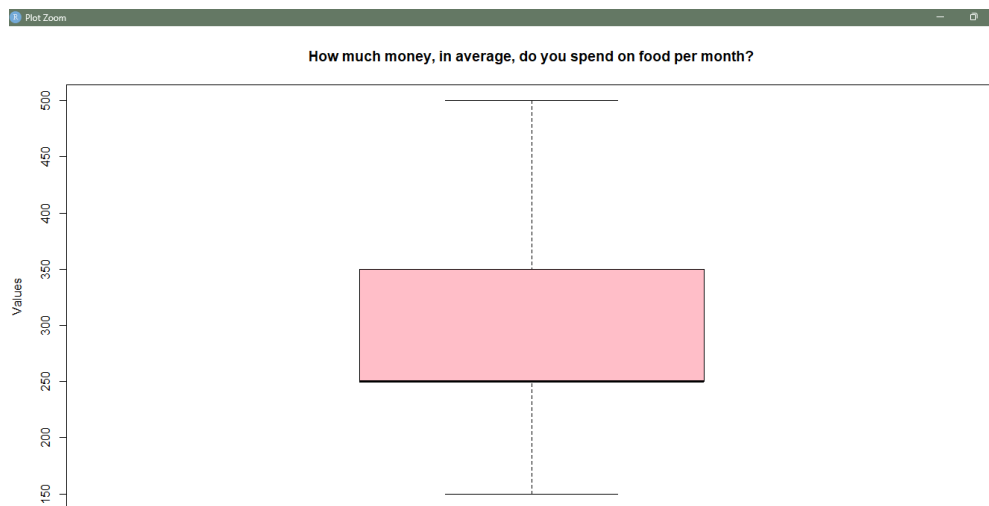
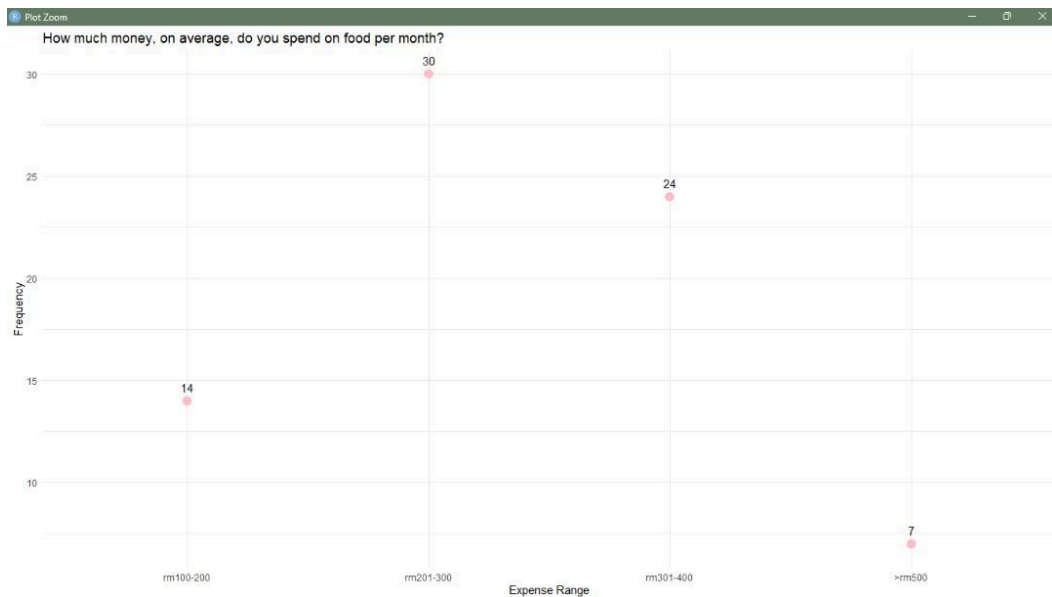
15. Average price of meal per day



```
summary(estimated_daily_expenses$Expense)
Min. 1st Qu. Median Mean 3rd Qu. Max.
 5.00  15.00  15.00 14.73  15.00 25.00
```

Explanation: There are 51 of the students who have spent on average of RM10-RM20 for their meals per day, followed by 13 of students who spent less than RM10 on their daily meals and lastly 11 students spent RM20-RM30 on average of meals taken per day.

16. Average price of meal per month



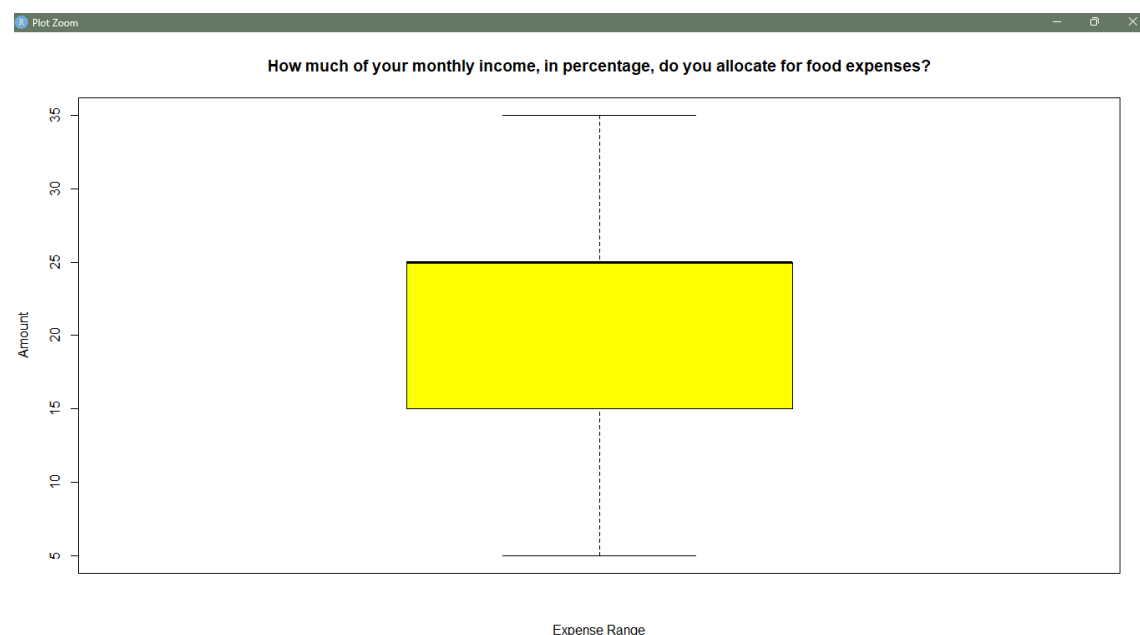
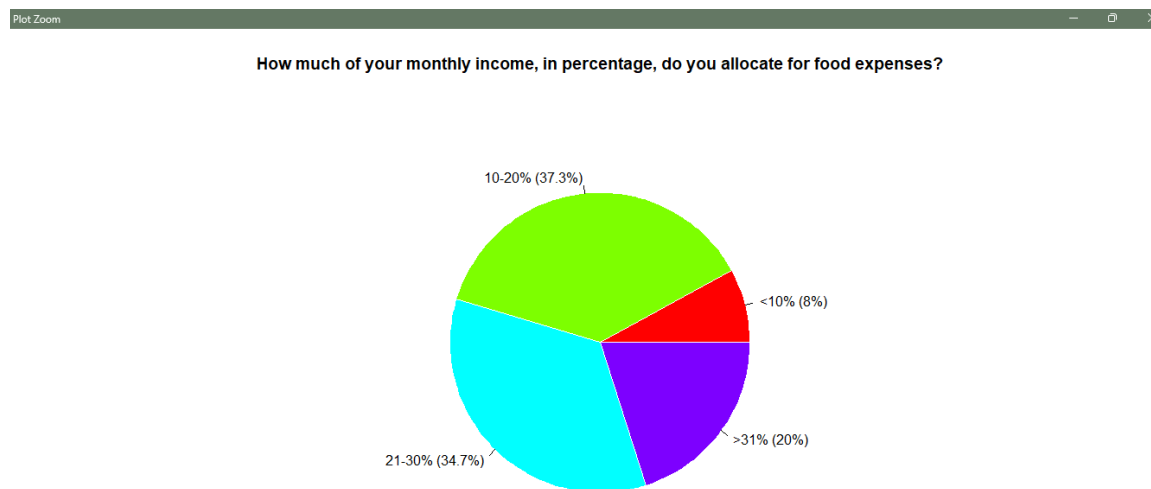
Average Money (midpoints)

```
summary(numeric_food_expenses$Expense)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
150.0	250.0	250.0	286.7	350.0	500.0

Explanation: For this section, students have to choose on the amount of money they spend on meals on average per month. From the dotplots graph, we can see that the majority of the students spent RM201-RM300 on average for their meal per month with 30 students. 24 of the students spent RM301-RM400 on average per month, followed by 14 students who spent RM100-RM200 on average per month and lastly, the least number of students with 7 spent more than RM500 on average per month.

17. Percentage of food expenses allocated from monthly expenses



```
summary(numeric_expense_data$Expense)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
5.00	15.00	25.00	21.67	25.00	35.00

Explanation: From the data, we found that the most percentages of expenses for food allocating from the monthly expenses is 10%-20% with 37% of the students followed by second most percentages of 35% of students which 21%-30%, 31% and above which is 20% of the students and then below 10% with 8% of the students.

4.0 CONCLUSION

In this study, 75 Malaysian university students including UTM and non-UTM students participated and most of them are 20 years old. Based on the analysis, we learned that the majority of the students take meals twice per day with a percentage of 49.3%. Their primary source and most preferable choice for the meal is by taking out from a restaurant. We also found out that most of the students will be dining out once to twice a week. The student rate for the frequency of their breakfast, lunch and dinner consumption from a scale of 1-5 is 1, 5 and 5 respectively. Moreover, the student will spend less than RM5 for breakfast and RM5-RM10 for lunch and dinner. On a scale from 1-5, most of the students rate 5 which is very important for the importance of budgeting for their food expenses. Majority of the students also rate 5 for the importance of nutrition in their meal choices. Lastly, students spent on average RM10-RM20 for food per day and RM201-300 per month. In addition, students will allocate around 10%-20% of their monthly expenses just for food expenses. As a conclusion, students rate nutrition and budgeting as extremely important, indicating that much focus is placed on these two areas. This data highlights the need for student-focused initiatives that encourage good eating practices and responsible money management. This study also teaches us how to use software to create and visualize the data we collected into different kinds of graphs, which helps greatly in our ability to analyze the data.

5.0 QUESTIONNAIRE

1. How old are you?
(Enter age)
2. Gender
 - Female
 - Male
3. Choose
 - UTM student
 - Non UTM student
4. How many meals do you eat per day?
 - 1
 - 2
 - 3
 - 4
 - 5
5. What is your primary source for meal?
 - Home-cooked
 - Take out
 - Dining out
6. How many times per week do you eat dining out?
 - 0
 - 1-2
 - 3-4
 - >4
7. On a scale of 1 to 5, how would you rate the frequency of your breakfast consumption?
 - Scale 1-5 (Rarely-Daily)
8. On a scale of 1 to 5, how would you rate the frequency of your lunch consumption?
 - Scale 1-5 (Rarely-Daily)
9. On a scale of 1 to 5, how would you rate the frequency of your dinner consumption?
 - Scale 1-5 (Rarely-Daily)

10. How much money do you typically spend on breakfast per meal?
 - <RM5
 - RM5-RM10
 - >RM10
11. How much money do you typically spend on lunch per meal?
 - <RM5
 - RM5-RM10
 - >RM10
12. How much do you typically spend on dinner per meal?
 - <RM5
 - RM5-RM10
 - >RM10
13. On a scale of 1 to 5, how would you rate the importance of budgeting for food expenses in your household?
 - Scale 1-5 (Not Important - Very Important)
14. How would you rank the importance of nutrition in your meal choices?
 - Scale 1-5 (Not Important - Very Important)
15. How much money, on average, do you spend on food per day?
 - <RM10
 - RM10<RM20
 - RM21<RM30
16. How much money, on average, do you spend on food per month?
 - RM100-RM200
 - RM201-RM300
 - RM301-RM400
 - >RM400
17. How much of your monthly income, in percentage, do you allocate for food expenses?
 - <10
 - 10-20
 - 21-30
 - >30

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Time	Address	Gender	How old are you?	Gender	Choose	How many meals Vital? Is your primary?	How many km On a scale 0 to 10	On a scale 0 to 10	On a scale 0 to 10	How much money do you have	How much do you type	How much do you type	On a scale 0 to 10	How would you rate it?	How much money do you have	On a scale 0 to 10	How much money do you have	On a scale 0 to 10	How much money do you have	
2	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
3	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
4	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
5	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
6	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
7	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
8	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
9	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
10	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
11	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
12	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
13	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
14	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
15	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
16	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
17	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
18	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
19	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
20	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
21	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
22	22/04/2024	amran@ad@ad	20 Female	Female	UTM student	3	Home-cooked	3-4	1	3	4	RMB-10	RMB-10	RMB-10	5	3	RMB-10	RMB-10	RMB-10	21-30	
23	22/04/2024	am																			

24